CBY-D06 OptiTrack/ROS2 setup and operation guide

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Overview

This document has been made to provide a guide to setting up, maintenance, and usage of the OptiTrack motion capture system in the CBY-D06 laboratory. As this is a working lab, keep in mind portions of this document may become outdated or obsolete over time. This document provides a quick guide to the current hardware setup, how to use the Motive software to track and capture motion data, and how to connect the tracking system with ROS2 for streaming applications.

Optitrack hardware setup

The motion capture system within D06 is currently configured within an enclosed cage environment, with an array of 20(twenty) Primex13 cameras. Each camera is connected and powered via a CAT6 PoE Ethernet connection. The camera angle and position can be manually adjusted using

motive pc setup

ensure the router above the white pc is plugged in and turned on. You should see the blinking status lights on the front of the router. Next, connect the red usb-c security key to the pc using the front port. Ensure the two indicator leds are lit. it is vital to ensure that this key is not lost or damaged, as it is required to use the optitrack motive software. Once connected, sign in to windows on the pc using your Uottawa network credentials.

Once signed in, launch the Optitrack Motive application found on the desktop or via the application panel on the start menu. Note this may take several seconds to launch. On launch you will see a message warning that the license has expired, this is known and not an issue at this time. After roughly 30s-1m, the motive software will launch. At this point, you should see the LED status rings around each optitrack camera shift to a solid green ring, confirming they have successfully connected to the software and are active. You can verify the camera connectivity and performance by switching to the Devices tab on the right side of the motive window.

With the motive software launched, you should now see UI elements on the screen. Firstly, the tracking area should be visible as a 3D environment in the center of the window. This shows the

computed position of each camera from the last calibration, as well as any currently detected Optitrack tracking markers or Assets. You can navigate and rotate this environment using the left and center mouse buttons, to rotate and move in 3D.

The Devices tab provides information and settings for the currently connected cameras. Specifically, this shows which cameras are connected, as well as the set frame rate, camera mode, and camera settings. Cameras can be individually disabled by toggling them from this menu, however doing so without recalibrating the system will impact overall performance.

ROS2 pc setup

note the following instructions have been tested on ROS2 Jazzy and ubuntu 24.04 LTS.

First install the optitrack ros2 repository found here :
https://github.com/MOCAP4ROS2-Project/mocap4ros2_optitrack.git

follow the given instructions to create a new workspace and build the packages. Once installed, open a new terminal window and copy the following commands:

1) `ros2 launch mocap4r2_optitrack_driver optitrack2.launch.py`
2) `ros2 lifecycle set /mocap4r2_optitrack_driver_node activate`

```
* at this point, you should see a message stating that the driver
has connected and found the motive pc. If you instead see a notice
that the system is not connected, verify the linux pc is on the
same network as the motive pc, and that the IP address set in the
configuration file matches the broadcasting IP in the
configuration panel in Motive.

To verify the driver is properly configured and running, you  can
use the following commmand in a new terminal:
```

3) `ros2 lifecycle set /mocap4r2_optitrack_driver_node activate`

communication and api usage

Depending on the application, you will most likely need to create your own script to extract the necessary data. An example script can be downloaded from this repository: https://github.com/garfield02-lgtm/Optitrack_ros2.git . This creates a basic publisher node that converts the motive raw data into a ROS2 PoseStamped message topic. Note that the example file applies an initial rotation and translation to the data to define the origin pose as the first observed pose.